Data Communication in Networks

Data communication refers to the process of transmitting digital or analog data between two or more devices through a communication medium.

Components of data communication include:

1. Sender (source)
2. Receiver (destination)
3. Transmission medium
4. Message (data)
5. Communication protocol

Examples:

- Email communication
- Video conferencing
- Online transactions
- File transfers

Passing Data Across a Network

When data moves across a network, it follows several steps:

1. Data is generated by the sending device
2. Data is divided into smaller units called **packets**
3. Packets travel through routers and switches
4. Packets are reassembled at the destination

Key technologies involved include:

- Packet switching
- Routing
- Error detection

Benefits of packet-based communication:

- Efficient bandwidth usage

- Reliable transmission
- Reduced congestion

Network Communication Models

Communication models describe **how data moves between devices in a network**.

1. The OSI Model

The **Open Systems Interconnection (OSI) Model** is a conceptual framework used to understand network communication.

Developed by the International Organization for Standardization.

It consists of **seven layers**:

1. Physical Layer – Transmits raw data bits
2. Data Link Layer – Handles node-to-node data transfer
3. Network Layer – Determines routing of packets
4. Transport Layer – Ensures reliable delivery
5. Session Layer – Manages communication sessions
6. Presentation Layer – Data formatting and encryption
7. Application Layer – User interface for network services

Advantages:

- Standardizes communication
- Simplifies troubleshooting
- Improves interoperability

2 TCP/IP Model

The **TCP/IP model** is the practical framework used on the internet.

Developed by the Defense Advanced Research Projects Agency.

It consists of four layers:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

Protocols used include:

- TCP
- IP
- HTTP
- FTP
- SMTP

Data Transmission Methods

Data can be transmitted across networks using different methods.

1 Simplex Transmission

Communication occurs in **one direction only**.

Example:

- Television broadcasting
- Radio transmission

Advantages:

- Simple implementation
- No interference from reverse signals

Disadvantages:

- No feedback from receiver

2 Half-Duplex Transmission

Communication occurs **in both directions but not simultaneously**.

Example:

- Walkie-talkies

Advantages:

- Allows two-way communication
- Efficient for moderate traffic

Disadvantages:

- Only one device can transmit at a time

3 Full-Duplex Transmission

Communication occurs **in both directions simultaneously**.

Example:

- Telephone communication
- Modern Ethernet networks

Advantages:

- Faster communication
- Reduced network collisions

Network Protocols and Standards

Network protocols are rules that govern how devices communicate over a network.

Protocols define:

- Data formatting
- Transmission procedures
- Error detection
- Synchronization

Examples include:

1. Transmission Control Protocol (TCP)

Ensures reliable delivery of data packets.

2. Internet Protocol (IP)

Handles addressing and routing of packets.

3. Hypertext Transfer Protocol (HTTP)

Used for web communication.

4. File Transfer Protocol (FTP)

Used for transferring files between computers.

5. Simple Mail Transfer Protocol (SMTP)

Used for sending email messages.